

Wall-Cleaning Robot

Strengths of the Proposal

The proposal addresses a genuine domestic safety problem: cleaning high walls and ceilings may require ladders, while mould, dust, and insects often accumulate in inaccessible areas. A robot capable of adhering to both vertical and inverted surfaces could reduce fall risk and repetitive manual work. The combination of suction, movable cleaning modules, orbital vibration, replaceable cloths, and debris filtration shows an attempt to integrate locomotion and cleaning into one physical system. The hand-drawn views also help communicate the mechanical concept.

Direct Critical Assessment

The proposal assumes suction will work on the exact surfaces most likely to need cleaning: dusty, porous, cracked, textured, or damp walls. Those are also the surfaces least able to maintain a vacuum seal. For a 3 kg robot with a safety factor of 3, the system must sustain at least 88 N before vibration and movement are considered. Even if 20 kPa pressure differential were achieved, the theoretical sealed area is about 0.0044 m²; leakage can destroy that holding force almost immediately. On a ceiling, one failed seal turns the cleaning product into a falling object. The design does not show independent suction chambers, pressure monitoring, a safety tether, edge detection, controlled descent, or a credible wall-to-ceiling transition. The cleaning claim is equally weak: vibrating mould with a cloth may disperse spores through the room, and wiping does not address the moisture source that caused the mould. Before discussing autonomous navigation or stain-recognition models, the team must prove the robot can remain attached without damaging paint or endangering people below.

Recommended Next Step

Narrow the first prototype to a tethered robot operating on a smooth vertical test board. Measure holding force, payload capacity, cleaning coverage, noise, energy consumption, and performance on several surface finishes. Once stable adhesion is demonstrated, add edge sensors and debris extraction before attempting ceiling operation.